TrES-5b

R-1106

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-5_150505` · created 2026-07-30T20:13:20+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457147.8389 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.185 +/- 0.054
Transit depth (Rp/Rs)^2	3.42 +/- 2.01 [%]
Orbital Inclination (inc)	89.57 +/- 2.86
Airmass coefficient 1 (a1)	1.62 +/- 1.12
Airmass coefficient 2 (a2)	-0.27 +/- 0.4
Scatter in the residuals of the lightcurve fit is	68.42 %
Best Comparison Star	#2 - [508, 60]
Optimal Aperture	2.27
Optimal Annulus	6.35
Transit Duration (day)	0.0875 +/- 0.0067

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.185 ± 0.054 vs 0.142 (deviation 0.71σ)
Duration fit vs published	0.0875 d vs 0.0758 d (deviation 1.54σ)
Mid-time O–C	2.9 min
Depth SNR	3.0
Residual scatter	68.42 %

Light curve

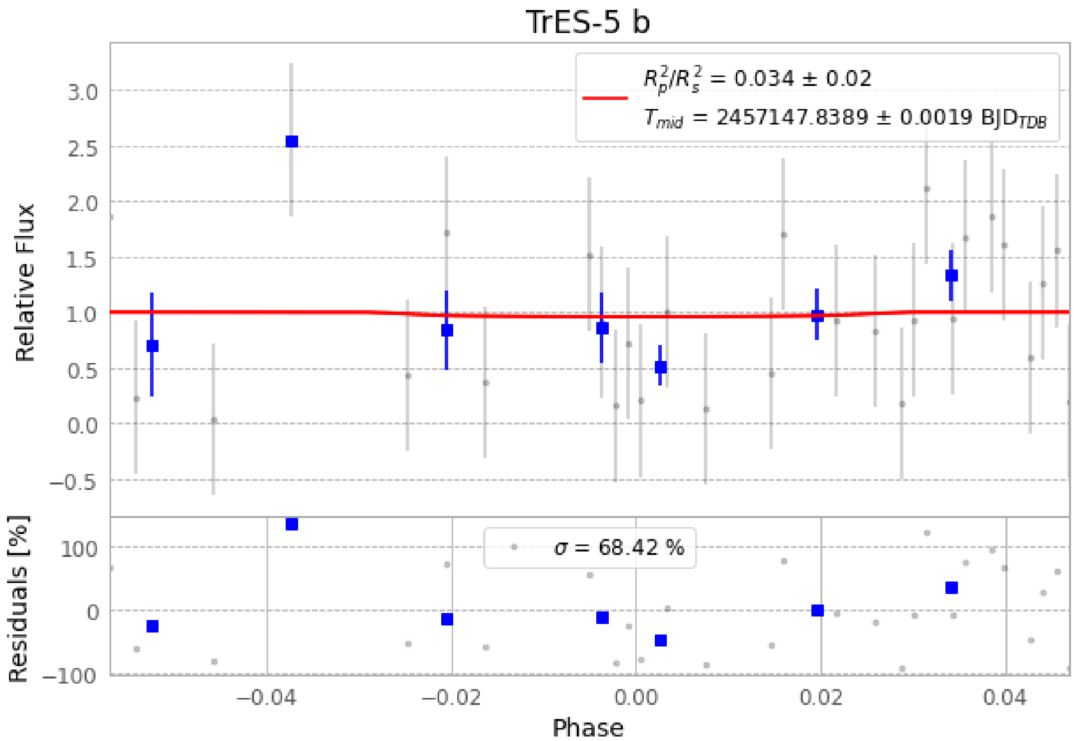

Figure 1 — FinalLightCurve_TrES-5 b_2015-05-04.png (SHA-256 2d86572ff3a06dee…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [154, 221], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 9a35f95bc99882f5…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

75 FITS frames (49 MB), `TRES-5150505060612.FITS` → `TRES-5150505095412.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-150505.log (90ae1265e30b…), FinalLightCurve_TrES-5 b_2015-05-04.png (2d86572ff3a0…), FinalLightCurve_TrES-5 b_2015-05-04.csv (29add2369362…)

Compute resources

Wall time 260.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)